

An Exact Capacity Criterion for Series with Eventual Congruences

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Abstract

Let Q_n be positive integers with $Q_n \mid Q_{n+1}$ and $Q_{n+1} \geq 2Q_n$, and let $F_n \geq 0$ be integer allowances with $\sum F_n/Q_n < \infty$. Series whose nonnegative integer digits satisfy these bounds eventually and are eventually divisible by every fixed integer represent an interval exactly when $Q_N \sum_{n>N} F_n/Q_n \rightarrow \infty$. In that case every nonnegative real is represented; otherwise the set of values is null and meagre. The sufficient direction also imposes eventual divisibility of the cumulative digit sums, with cutoffs common to every target in the interval. Arbitrary oscillation, zero allowances and unbounded denominator ratios are permitted. For factorial denominators and allowances n^c on a fixed support, the criterion gives the sharp gap threshold $\text{gap} < c$; its contrast with the unrestricted threshold $\text{gap} \leq c$ occurs exactly at integer exponents. The proof combines common continuation intervals for every cumulative residue with the arithmetic lattice of prefixes. A separate common-divisor criterion isolates an obstruction to rationality for bounded-carry Cantor series.

1 The exact criterion

A factorial expansion usually permits a digit of size about n at position n . If more digits are allowed, can most positions be discarded? What changes if each fixed integer must eventually divide every digit?

Let Q_n be positive integers, $n \geq 1$, such that $Q_n \mid Q_{n+1}$ and $Q_{n+1} \geq 2Q_n$. Let $F_n \in \mathbb{Z}_{\geq 0}$ satisfy $\sum_n F_n/Q_n < \infty$, and put

$$U_N = Q_N \sum_{n>N} \frac{F_n}{Q_n}. \tag{C}$$

Define $E_Q(F)$ to be the set of sums $\sum_{n \geq 1} e_n/Q_n$, allowing nonnegative integer digits $e_n \leq F_n$ eventually and requiring, for each fixed $q \geq 1$, that $q \mid e_n$ eventually. The cutoffs and finite initial digits may depend on the represented value.

Theorem 1. *If $U_N \rightarrow \infty$, then $E_Q(F) = [0, \infty)$; otherwise $E_Q(F)$ is null and meagre. In particular interval filling is equivalent to $U_N \rightarrow \infty$. In the positive case, every nonnegative target may be represented with $e_n = 0$ before any prescribed cutoff, at most one exception to $e_n \leq F_n$, and*

$$q \mid e_n, \quad q \mid \sum_{k < n} e_k \quad \text{for all sufficiently large } n,$$

with congruence cutoffs depending on q but independent of the target. If the allowance must hold at every position, a nondegenerate interval is still represented with both congruences and a common cutoff for each q .

The criterion compares *all* future capacity with the prefix lattice. It requires no monotonicity, regular variation or gap bound on the allowances, and no upper bound on the denominator ratios. It does not require the Q_n to clear every rational denominator. Section 2 proves the theorem using the same residue-feedback mechanism already present in the public Lean repository. The capacity estimates and complete criterion are ordinary proofs; the abstract feedback endpoint is kernel-checked separately. Both geometric and factorial denominators satisfy the hypotheses.

1.1 The sharp support threshold

Let $S \subseteq \mathbb{N}_{\geq 0}$ and $c > 0$. Define $E(S, c)$ to be the set of sums

$$x = \sum_{n \geq 1} \frac{e_n}{n!}, \quad e_n \in \mathbb{Z}_{\geq 0}, \quad e_n = 0 \ (n \notin S), \quad e_n \leq n^c \text{ eventually}, \quad (1)$$

subject to

$$\text{for every } q \geq 1, \quad q \mid e_n \text{ for all sufficiently large } n. \quad (2)$$

The cutoffs and the finite initial digits in this definition may depend on x . All series converge, since finitely many unrestricted digits do not affect convergence. A permitted position need not carry a nonzero digit.

Theorem 2. *If S is finite, $E(S, c)$ is countable. If S is infinite, enumerate it as $n_0 < n_1 < \dots$.*

1. *For $0 < c \leq 1$, $E(S, c)$ is null and meagre.*
2. *For $c > 1$, $E(S, c)$ contains a nondegenerate interval if and only if*

$$n_j - n_{j-1} < c \quad \text{for all sufficiently large } j. \quad (3)$$

If this condition fails, $E(S, c)$ is null and meagre.

When (3) holds, one may require $e_n = 0$ before any prescribed cutoff and impose both

$$q \mid e_n, \quad q \mid \sum_{k < n} e_k \quad (4)$$

for all sufficiently large n , with cutoffs depending on q but independent of x throughout the constructed interval.

The theorem is an ordinary mathematical proof. The associated formal sources check the general digit-feedback construction and, separately, the common-divisor carry obstruction of Section 6; they do not formalise this gap classification or its measure argument. The source and attribution account is in Section 7.

For example, with allowance n^2 , every second position suffices without congruences. Under (2), infinitely many omitted positions already force a null set. With allowance $n^{2.01}$, every second position again suffices, even under (4). Thus the strict inequality in (3) is essential.

Corollary 3. *For $c > 1$, the least possible asymptotic density of a fixed permitted support that fills an interval under (2) is $1/(\lceil c \rceil - 1)$. Every such support has lower density at least this value, and an arithmetic progression attains it. A density bound alone is not sufficient: even rare gaps of length $\lceil c \rceil$ prevent interval filling if they occur infinitely often.*

Proof. Put $d = \lceil c \rceil - 1$. Eventual gaps at most d give $\liminf_{N \rightarrow \infty} |S \cap [1, N]|/N \geq 1/d$. An arithmetic progression of step d satisfies the theorem. $\square$

2 Proof of the capacity criterion

The necessity and sufficiency use different parts of the expansion. Prefixes supply an arithmetic obstruction; a common interval of possible future remainders makes the residue choices compatible with exact representation.

2.1 Bounded capacity along a subsequence

If U_N does not tend to infinity, there are a constant $B > 0$ and arbitrarily large N with $U_N \leq B$. Choose one integer $q > B$. Fix a cutoff K after which the allowance and divisibility by q hold, and fix the finite prefix, of value y_0 . The set V of resulting sums is compact by summability and the finite choices at each remaining position. For every such $N > K$,

$$V \subseteq y_0 + \frac{q}{Q_N} \mathbb{Z} + [0, B/Q_N].$$

Indeed the scaled prefix after K is a multiple of q and the remaining tail is at most U_N/Q_N . These grids leave holes inside every fixed open interval for all sufficiently large selected N , so V is nowhere dense. For a bounded interval J they also give $|V \cap J| \leq (B/q)|J| + 2B/Q_N$. Passing to the subsequence and covering V by finitely many intervals with total length at most $|V| + \varepsilon$ gives $|V| \leq (B/q)(|V| + \varepsilon)$, hence $|V| = 0$. A countable union over the cutoffs and integer prefixes covers $E_Q(F)$.

2.2 One continuation interval for every cumulative residue

Suppose $U_N \rightarrow \infty$ and put $h_N = \inf_{k \geq N} U_k$. For sufficiently large n , let M_n be the largest factorial at most $\min\{n, \sqrt{h_{\lfloor n/2 \rfloor}}\}$, and set the finitely many earlier moduli equal to one. These positive integers are nested under divisibility, tend to infinity, and are eventually divisible by each fixed integer. The delayed lower envelope controls the whole future cost of rounding, even when the denominator ratios are unbounded. Indeed, $Q_N/Q_k \leq 2^{N-k}$, so for sufficiently large N ,

$$\begin{aligned} C_N := Q_N \sum_{k > N} \frac{M_k}{Q_k} &\leq \sqrt{h_N} \sum_{k=N+1}^{2N} 2^{N-k} + \sum_{k > 2N} k 2^{N-k} \\ &\leq \sqrt{h_N} + (2N+2)2^{-N}. \end{aligned}$$

The first bound uses $\lfloor k/2 \rfloor \leq N$ when $k \leq 2N$; the second uses $M_k \leq k$. Since $U_N \geq h_N \rightarrow \infty$ and $M_N \leq \sqrt{h_N}$, this proves $U_N - 2C_N \geq M_N$ for every sufficiently large N . Start beyond this point and any prescribed cutoff.

Keep only the active positions $S = \{n : F_n \geq 2M_n\}$ after this cutoff. Define two common tail bounds

$$\alpha_N = \sum_{\substack{n > N \\ n \in S}} \frac{M_n}{Q_n}, \quad \beta_N = \sum_{\substack{n > N \\ n \in S}} \frac{F_n - M_n}{Q_n}.$$

Both are nonnegative and tend to zero. Discarding an inactive position loses less than $2M_n$; reserving both margins at an active position loses exactly $2M_n$. Thus the preceding estimate gives the decisive overlap inequality

$$Q_N(\beta_N - \alpha_N) \geq U_N - 2C_N \geq M_N. \quad (\text{O})$$

In particular the active set is infinite and each continuation interval has positive width.

Choose a target in $[\alpha_{N_0}, \beta_{N_0}]$ and set all preceding digits to zero. Suppose the current unweighted cumulative sum is C and the remaining target before position n lies in $[\alpha_{n-1}, \beta_{n-1}]$.

At an inactive position choose zero. At an active position, the permitted digits in the residue class $e \equiv -C \pmod{M_n}$, $0 \leq e \leq F_n$, form a nonempty arithmetic progression of step M_n . Its least digit is less than M_n and its greatest digit exceeds $F_n - M_n$. By (O), the intervals

$$\frac{e}{Q_n} + [\alpha_n, \beta_n]$$

for these digits overlap. Their union contains

$$\left[\frac{M_n}{Q_n} + \alpha_n, \frac{F_n - M_n}{Q_n} + \beta_n \right] = [\alpha_{n-1}, \beta_{n-1}].$$

Choose a digit whose interval contains the remaining target. The next remainder stays between α_n and β_n , and these bounds tend to zero, proving exact representation. This is precisely the common-interval feedback construction; no target-independent ordinary block sum is required.

After an active position n , the cumulative sum is divisible by M_n . Each later active digit is divisible by the previous active modulus, because both adjacent cumulative sums are; the moduli are nested. For a fixed q , take one active position with $q \mid M_n$. After that position, all individual digits and all preceding cumulative sums are divisible by q . This cutoff is common to all targets in the interval.

2.3 From one interval to every nonnegative target

The finite-exception convention in $E_Q(F)$ strengthens the conclusion. Given $y > 0$, choose N arbitrarily late with $\alpha_N < y$ and put

$$p = M_N \left\lfloor \frac{Q_N(y - \alpha_N)}{M_N} \right\rfloor.$$

Then $p \geq 0$, $M_N \mid p$, and (O) gives $y - p/Q_N \in [\alpha_N, \beta_N]$. Set $e_N = p$ and all earlier digits to zero. Run the same continuation construction after N , with initial cumulative sum $C = p$; its interval covering worked for every C . All later digits obey their allowances. The zero target uses zero digits.

The congruence cutoffs can also be common to all y . For fixed q , choose an active position j with $q \mid M_j$. If the target's exceptional index N precedes j , the repair at j gives the required divisibilities after j . If $N \geq j$, all preceding digits are zero, $q \mid M_N \mid p$, and all subsequent repairs preserve divisibility by q . Thus $j + 1$ is a valid cutoff for every target. The allowance-exception index itself may depend on the target. This completes Theorem 1.

The growth hypothesis has content. If repeated denominators are allowed, take $Q_n = 2^{\lfloor \sqrt{n} \rfloor}$ and $F_n = 1$. The allowance series converges, while $U_N \geq \lfloor \sqrt{N} \rfloor$ by counting the next complete denominator block. Yet eventual divisibility by two forces all sufficiently late digits to vanish, so the attainable set is countable. Imposing all cumulative congruences as well leaves only zero. Thus nestedness alone does not justify the criterion.

3 Why a large gap prevents interval filling

The decisive obstruction is local. At a position N , the whole preceding factorial sum lies on a lattice of spacing $1/N!$. Eventual divisibility widens that spacing to $q/N!$, up to a fixed translation. After a sufficiently long gap, all remaining permitted digits reach only a bounded multiple of $1/N!$. Choosing one fixed q larger than that bound leaves holes at arbitrarily small scales.

Here are the details, including the target-dependent cutoffs in (1)–(2). Put $r = \lceil c \rceil$. Suppose infinitely many successive S -gaps have length at least r , and let N run through the positions immediately before these gaps. For large k ,

$$\frac{(k+1)^c/(k+1)!}{k^c/k!} = \frac{(1+1/k)^c}{k+1} \leq \frac{1}{2}.$$

Consequently, for sufficiently large such N ,

$$N! \sum_{\substack{k > N \\ k \in S}} \frac{k^c}{k!} \leq 2N! \frac{(N+r)^c}{(N+r)!} \quad (1)$$

$$\leq 2^{c+1} N^{c-r} \leq B, \quad B = 2^{c+1}. \quad (5)$$

In the second inequality we used $N \geq r$ and $(N+1) \cdots (N+r) \geq N^r$.

The bounded capacity along this subsequence invokes the necessity argument of Theorem 1, with allowances $F_n = \lfloor n^c \rfloor$ on S and zero elsewhere. It gives nullity and meagreness even with target-dependent congruence and allowance cutoffs.

If $c \leq 1$, every support gap is at least $r = 1$, so this proves the first part of Theorem 2. If $c > 1$ and (3) fails, it proves the negative part. Notice that infrequent large gaps are enough; average digit counts do not detect this obstruction.

4 A digit that also corrects the cumulative residue

The following construction supplies the positive direction. It is useful beyond factorial weights.

Lemma 4. *Let $w_j > 0$, let positive integers M_j satisfy $M_{j-1} \mid M_j$, and let $F_j \geq 0$. Suppose $M_j w_j \rightarrow 0$ and, for $j \geq 1$,*

$$2M_j \leq M_{j-1} \frac{w_{j-1}}{w_j}, \quad 2M_{j-1} \frac{w_{j-1}}{w_j} \leq F_j. \quad (7)$$

Every $y \in [M_0 w_0, 2M_0 w_0]$ has a representation $y = \sum_{j \geq 1} b_j w_j$ with integers $0 \leq b_j \leq F_j$ such that

$$M_j \mid \sum_{i \leq j} b_i, \quad M_{j-1} \mid b_j.$$

Proof. Start with residual $R_0 = y$ and cumulative sum $C_0 = 0$. Suppose $M_{j-1} w_{j-1} \leq R_{j-1} \leq 2M_{j-1} w_{j-1}$ and $M_{j-1} \mid C_{j-1}$. Put $u = R_{j-1}/w_j$ and let v be the least nonnegative residue of $-C_{j-1}$ modulo M_j . Choose

$$b_j = v + M_j \left\lfloor \frac{u - M_j - v}{M_j} \right\rfloor. \quad (8)$$

Since $u \geq 2M_j$ and $0 \leq v < M_j$, this integer is nonnegative. The floor identity gives $M_j \leq u - b_j < 2M_j$; also $b_j \leq u \leq F_j$. Thus $R_j = R_{j-1} - b_j w_j$ lies in $[M_j w_j, 2M_j w_j]$, while $C_j = C_{j-1} + b_j$ is divisible by M_j . The incoming modulus divides both C_{j-1} and M_j , hence also b_j . Finally $R_j \rightarrow 0$, so the partial sums converge to y . $\square$

The residue in (8) may depend on y . Only the schedule of moduli needs to be common to all targets. Requiring target-independent ordinary block sums would impose an unnecessary restriction on the construction.

5 The positive direction and the arithmetic cost

Suppose $c > 1$ and the support gaps are eventually at most $d = \lceil c \rceil - 1$. Discard a finite prefix and write

$$r_j = \frac{n_j!}{n_{j-1}!}, \quad F_j = n_j^c, \quad w_j = \frac{1}{n_j!}.$$

Then $r_j \geq n_j \rightarrow \infty$, while $r_j \leq n_j^d$ and therefore $F_j/r_j \geq n_j^{c-d} \rightarrow \infty$. Define

$$h_j = \inf_{k \geq j} \min \left\{ \frac{\sqrt{r_k}}{2}, \frac{F_k}{2r_k} \right\}.$$

This lower envelope is nondecreasing and tends to infinity. Begin sufficiently late that $h_1 \geq 1$, set $M_0 = 1$, and let M_j be the largest factorial at most h_j . These moduli are nested and eventually divisible by every fixed integer. Moreover,

$$2M_j \leq \sqrt{r_j} \leq r_j M_{j-1}, \quad 2r_j M_{j-1} \leq F_j.$$

The second inequality uses $M_{j-1} \leq h_j$, also valid for $j = 1$. Finally, $M_j/n_j! \leq \sqrt{r_j}/(2n_j!) \leq 1/(2\sqrt{n_j!}) \rightarrow 0$. Lemma 4 fills $[1/n_0!, 2/n_0!]$ using the positions n_j , $j \geq 1$. Put zero digits elsewhere. Once $q \mid M_j$, all later digits and cumulative sums at the original integer indices have the divisibilities in (4). This proves the positive direction, including common cutoffs.

For comparison, remove condition (2) and call the resulting attainable set $E_0(S, c)$.

Proposition 5. *For an infinite support and $c \geq 1$, $E_0(S, c)$ contains an interval exactly when its successive gaps are eventually at most $\lceil c \rceil$. Otherwise it is null and meagre. For $0 < c < 1$ it is always null and meagre.*

Proof. For necessity repeat Section 3 with $r = \lfloor c \rfloor + 1$. The bound in (5) now tends to zero. With lattice spacing $1/N!$, it is eventually bounded by $1/(2N!)$, so the same compact-set argument applies with $q = 1$, $B = 1/2$. For sufficiency, $r_j \leq n_j^c$ along the support. The ordinary mixed-radix expansion using digits $0 \leq b_j < r_j$ fills $[0, 1/n_0!]$. To see this directly, the maximum tail after n_j is $1/n_j!$, since $(r_k - 1)/n_k! = 1/n_{k-1}! - 1/n_k!$. The adjacent digit intervals therefore meet, and their lengths tend to zero. $\square$

Thus the difference between the two gap thresholds occurs exactly at integer c . At $c = m \geq 2$, imposing eventual congruences raises the least support density from $1/m$ to $1/(m-1)$. At $c = 1$ it destroys interval filling entirely. With quadratic allowances, deleting only the positions 2^k destroys interval filling, although the permitted support still has density one. This is the simplest example of why support density loses decisive information. These statements concern intervals of values on a common permitted support; they are not lower bounds for representing one specially chosen value.

6 A common-divisor test for irrationality

The distinction between small and large allowances also appears directly in rationality. Let $b_n \geq 2$ be integers, put $Q_0 = 1$ and $Q_n = b_1 \cdots b_n$, and consider a Cantor series.

Theorem 6. *Suppose $0 \leq a_n \leq A$ and $e_n \geq 0$ are integers, $e_n \leq Cb_n$ eventually, and $a_n + e_n$ is nonzero infinitely often. If*

$$\limsup_{n \rightarrow \infty} \gcd(b_n, e_n) = \infty,$$

then $\sum_{n \geq 1} (a_n + e_n)/Q_n$ is irrational.

Proof. For large n , the scaled tail satisfies

$$0 < Q_n \sum_{k>n} \frac{a_k + e_k}{Q_k} \leq A + 2C.$$

Indeed, the bounded a_k contribute at most $A \sum_{j \geq 1} 2^{-j} = A$, and $e_k \leq Cb_k$ contributes at most $C \sum_{j \geq 1} 2^{-(j-1)} = 2C$. If the total were p/D , the numbers

$$T_n = DQ_n \left(\frac{p}{D} - \sum_{k \leq n} \frac{a_k + e_k}{Q_k} \right)$$

would be positive integers eventually bounded by $D(A + 2C)$. No hypothesis that $D \mid Q_n$ is needed: the extra factor D clears it. The recurrence

$$b_n T_{n-1} = D(a_n + e_n) + T_n$$

implies $\gcd(b_n, e_n) \mid Da_n + T_n$. The integer on the right is positive and eventually at most $D(2A + 2C)$, contradicting the unbounded gcd. $\square$

For factorial denominators, $b_n = n$, and eventual divisibility of e_n by each fixed integer makes $\gcd(n, e_n)$ arbitrarily large arbitrarily late: choose a late multiple of that integer. Hence bounded nonnegative digits that are nonzero infinitely often cannot be rationalised by nonnegative $O(n)$ corrections satisfying (2). Conversely any allowance $F(n)$ with $F(n)/n \rightarrow \infty$ permits interval filling under (4): apply the positive construction with $r_n = n$, replacing n^c by $F(n)$.

The arithmetic hypothesis cannot be replaced by $b_n \rightarrow \infty$, even if both congruences (4) are retained. Set $a_n = 1$, $e_1 = 2$, and $e_n = (n+1)! - n!$ for $n \geq 2$, and let $b_n = e_n + 2$. Then $\sum_{k \leq n} e_k = (n+1)!$, so (4) holds, and $e_n < b_n$. Nevertheless

$$\sum_{n \geq 1} \frac{a_n + e_n}{Q_n} = \sum_{n \geq 1} \frac{b_n - 1}{Q_n} = 1.$$

Here $\gcd(b_n, e_n) \leq 2$. This example separates rapid denominator growth from the arithmetic obstruction used in Theorem 6.

7 Sources, formal correspondence, and further questions

The starting point is the sparse perturbation construction accompanying Erdős Problem #251 in this repository, especially its [short paper](#) and [ResidueFeedbackCore.lean](#). The latter already proves residue-dependent selection and an abstract infinite sum endpoint. An operator-supplied review supplied the form of Lemma 4, the sharp exponential support constants, and the linear/superlinear factorial contrast. Those ingredients are credited to that review, not presented as discoveries of this note. The extensions developed here are the exact capacity criterion on arbitrary strict integer divisibility chains, the factorial support classification and its integer-exponent comparison, and the common-divisor formulation.

Airey, Mance and Vandehey already use digit sets eventually divisible by every fixed integer while retaining asymptotically full digit entropy [1, Section 6, p. 1321]. Their theorem concerns normality and Hausdorff dimension for chosen Cantor bases. Here a fixed divisibility chain and arbitrary summable allowances are given, and the conclusion distinguishes interval filling from nullity and meagreness. These are elementary arguments in the classical theory of Cantor series and achievement sets; historical novelty of the exact classification is not

established. Classical interval covering is background, rather than a contribution claimed here. A literature comparison and the reproducible checks are recorded in [the accompanying research record](#). The work was developed with AI assistance and mathematical cross-checking by separate agent passes; that does not constitute independent expert review.

The formal module `CongruenceInterpolation.lean` uses the existing feedback module and states the common-divisor obstruction for a real carry recurrence. The analytic identification of that recurrence with the Cantor series, and the capacity and support classifications, are the ordinary proofs above. The module `FeedbackContinuation.lean` reuses the existing interval-feedback endpoint and proves eventual individual and cumulative divisibility from nested cofinal moduli; choosing the moduli and continuation intervals remains part of the ordinary proof. See the research record for the exact build status and source revision. No claim about any of the eight Erdős programmes changes; in particular factorial denominators $n!$ here are not $n! - 1$ from #68.

The capacity criterion already covers non-power and oscillating allowances. For factorial gaps of fixed length m , a bounded multiple of n^m still gives the lattice obstruction, whereas $n^m L(n)$ with $L(n) \rightarrow \infty$ permits interval filling. A further question concerns the null case: what finer tail data determine its Hausdorff dimension? The criterion itself does not separate dimension zero from full-dimensional null sets. Outside integer divisibility chains the prefix lattice changes, so no corresponding necessity is asserted here.

References

- [1] D. Airey, B. Mance and J. Vandehey, *Normality preserving operations for Cantor series expansions and associated fractals. II*, New York J. Math. **21** (2015), 1311–1326. <https://nyjm.albany.edu/j/2015/21-60v.pdf>.